

Assignment no 1.

Variational Quantum Circuits for ground state properties of the XXZ model

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This work studies the finite size L spin-1/2 chain described by the XXZ Hamiltonian via a Variational Quantum Circuit approach. Our investigation can be structured in 4 different parts. Firstly, our main goal is to investigate the ground state properties. Secondly, the time evolution of this state. Then, we also explore the properties of the First Excited State using this same methodology and finally we take an insight into the degeneracy of the system. During the whole study we propose three different ansatzes based on different characteristics, and our results highlight the importance of using a circuit design tailored to the specific Hamiltonian.

I. INTRODUCTION

Simulating quantum many-body systems is fundamentally limited by the curse of dimensionality, where the Hilbert space grows exponentially with system size. Storing the wave function of a quantum many-body system requires 2^N complex amplitudes; for $N \approx 50$, full state-vector simulation pushes the memory limits of even the world's first exascale supercomputers [1].

This computational bottleneck motivated the development of quantum computation, a concept famously crystallized by Feynman: “Nature isn’t classical, dammit, and if you want to make a simulation of nature, you’d better make it quantum mechanical” [2]. Despite the immense potential, fault-tolerant quantum computers are not yet a reality. We currently operate in the Noisy Intermediate-Scale Quantum (NISQ) era, characterized by processors where limited qubit counts and noise prevent the execution of deep circuits [3].

Since fault-tolerant algorithms are currently out of reach, Variational Quantum Algorithms (VQAs) have emerged as the leading candidates for demonstrating near-term quantum advantage [4]. These hybrid algorithms offload computational complexity to classical optimizers, keeping the quantum circuit depth manageable. The Variational Quantum Eigensolver (VQE) serves as the standard benchmark of quantum chemistry and condensed matter simulation. However, significant challenges remain, most notably the presence of barren plateaus [5] and the cost of shot noise [6].

This project focuses on the Heisenberg XXZ spin-1/2 chain, a cornerstone model in condensed matter physics. As the XXZ model is integrable via the Bethe Ansatz [7], it serves as an ideal testbed for benchmarking quantum algorithms against exact results [8, 9]. Our objectives are threefold. Firstly, we will characterize the ground state energy and magnetic order using VQE [10]. We will then simulate the time-evolution of the system. Finally, we will extend the study to excited states using VQD [11] to observe spectral properties such as Kramers’ degeneracy.

II. PROBLEM FORMULATION

The primary objective of this project is to study finite size L spin-1/2 chain described by the *XXZ Hamiltonian*:

$$\hat{H} = - \sum_i \hat{X}_i \hat{X}_{i+1} + \hat{Y}_i \hat{Y}_{i+1} + \delta \hat{Z}_i \hat{Z}_{i+1}, \quad (1)$$

where $\delta \in [-2, 2] \subset \mathbb{R}$ is the *anisotropy parameter*.

Physically, this Hamiltonian can be decomposed into two distinct contributions: the *hopping term* ($\hat{X}_i \hat{X}_{i+1} + \hat{Y}_i \hat{Y}_{i+1}$), which allows spin excitations to move between neighbours, and the *interaction term* ($\delta \hat{Z}_i \hat{Z}_{i+1}$), which energetically favours or penalizes aligned spins depending on the sign of δ .

This parameter δ plays a critical role in the system’s phase diagram. The point $\delta = 1$ corresponds to the isotropic Heisenberg model, marking a quantum phase transition between the gapless XY phase ($|\delta| < 1$) and the gapped ferromagnetic Ising phase ($\delta > 1$). This transition has significant consequences for simulation:

- **Entanglement:** At the critical point ($\delta = 1$), the system exhibits long-range quantum correlations and maximum entanglement entropy.
- **Complexity:** In gapped phases (e.g., $\delta \approx 2$), the ground state approximates a product state, which is relatively simple to represent.

We investigate this system using a *Variational Quantum Circuit* (VQC) approach. This method employs a parameterized model: specifically, a quantum circuit constructed from rotation gates with tunable parameters θ . We propose three distinct circuit designs with varying degrees of connectivity and gate types: the Hardware Efficient, the Hamiltonian-Inspired and the All-to-All design. Our analysis is structured as follows:

A. Exact Diagonalization (ED) (Benchmarking):

We first compute the exact ground state energy, entanglement entropy, and Pauli expectation values/variances using classical matrix diagonalization. This provides the “ground truth” against which we validate our quantum models.

B. Variational Quantum Circuits (VQC): We optimize parameterized quantum circuits to tackle four distinct tasks:

1. **Ground State Search (VQE):** Minimizing energy to find the system's vacuum state.
2. **Time Evolution:** Simulating the dynamic evolution $|\psi(t)\rangle = e^{-iHt}|\psi(0)\rangle$.
3. **First Excited States (VQD):** Modifying the loss function to find the minimum energy state which is orthogonal to the ground state.
4. **Kramers' Degeneracy:** Repeat last part with special attention to symmetry-induced degeneracies.

III. METHODS

This section outlines the computational frameworks employed to investigate the XXZ spin chain. We begin by defining the ED approach, which serves as the classical benchmark for our results. Subsequently, we detail the VQC architectures and the optimization strategies used to approximate the ground state, simulate time evolution and finding the first excited state.

A. Exact Diagonalisation

To evaluate the performance of our quantum algorithms, we first compute the exact solutions of the system. While ED yields the precise eigenvalues and eigenvectors of the Hamiltonian, it is computationally expensive; the dimension of the Hilbert space scales exponentially as 2^L . For a system of size L , the Hamiltonian $\hat{H}$ is a matrix of dimension $2^L \times 2^L$. For the system sizes considered in this work ($L \in [2, 10]$), this approach remains computationally tractable and provides the necessary baseline for benchmarking

The procedure for ED is as follows:

1. **Matrix Construction:** We construct the Hamiltonian matrix in the computational basis. For $L = 10$, this corresponds to a 1024×1024 complex matrix.
2. **Eigendecomposition:** We diagonalize the matrix to obtain the full spectrum of eigenvalues $\{E_n\}$ and eigenstates $\{|\psi_n\rangle\}$.
3. **Ground State Selection:** We identify the ground state $|\psi_0\rangle$ corresponding to the minimum energy eigenvalue E_0 . This state serves as the target for our VQC ansatz optimization.

To characterize the phase diagram of the XXZ model and validate the VQC results, we verify the following physical quantities as a function of the anisotropy $\delta \in [-2, 2]$.

a. Ground State Energy. We find the value for the ground state energy depending on the anisotropy parameter, i.e. we track $E_0(\delta)$.

b. Bipartite Entanglement Entropy. To quantify quantum correlations, we compute the bipartite entanglement entropy [12]. The spin chain is partitioned into two subsystems of equal size, A and B . For a pure state $|\psi\rangle_{AB}$, the entanglement is given by the von Neumann entropy of the reduced density matrix of subsystem A , denoted as $\rho_A = \text{Tr}_B(|\psi\rangle\langle\psi|)$:

$$E(\psi) = S(\rho_A) = -\text{Tr}(\rho_A \ln \rho_A).$$

The value of $E(\psi)$ provides insight into the nature of the state:

- $E(\psi) = 0$: The system is in a product state $|\psi\rangle = |a\rangle \otimes |b\rangle$, indicating no entanglement (separable).
- $E(\psi) > 0$: The subsystems are entangled. High entropy indicates strong long-range quantum correlations, which are characteristic of quantum critical points (e.g., at $\delta = 1$).

The entropy is bounded by the dimension of the subsystems:

$$0 \leq E(\psi) \leq \log \min(d_A, d_B), \quad (2)$$

where d_A and d_B denote the Hilbert space dimensions of subsystems A and B , respectively.

c. Magnetization and Variances. We analyze the system's magnetic properties by computing the expectation values and variances of the total spin operators $\hat{\tau}^\alpha = \sum_i \hat{\sigma}_i^\alpha$ for $\alpha \in \{x, y, z\}$.

The expectation values $\langle \hat{\tau}^\alpha \rangle$ represent the net magnetization along a specific axis. Due to the $U(1)$ symmetry of the XXZ Hamiltonian [13], [14] (invariance under rotation around the z -axis), we expect the transverse magnetizations to vanish: $\langle \hat{X} \rangle = \langle \hat{Y} \rangle = 0$.

Conversely, the longitudinal magnetization $\langle \hat{Z} \rangle$ is expected to vary with δ and L , reflecting the alignment of spins in the Ising phases or the anti-alignment in the antiferromagnetic regime.

We also compute the variances $\langle \Delta^2 \hat{\tau}^\alpha \rangle = \langle (\hat{\tau}^\alpha)^2 \rangle - \langle \hat{\tau}^\alpha \rangle^2$ to assess quantum fluctuations. It is important to remark that the $U(1)$ symmetry of the XXZ Hamiltonian implies the conservation of total magnetization: $[\hat{H}, \hat{Z}] = 0$. Consequently, the exact ground state is an eigenstate of $\hat{Z}$, which theoretically mandates a zero variance $\langle \Delta^2 Z \rangle = 0$ across the entire phase diagram.

All these properties will be plotted against the value of the anisotropy parameter δ and for different sizes of the system L .

B. Variational Quantum Circuits

While Exact Diagonalization provides a precise benchmark, it scales exponentially with system size. To address this, we employ Variational Quantum Algorithms (VQAs), which utilize a parameterized quantum circuit as a function approximator. The quantum state is represented as $|\psi(\theta)\rangle = U(\theta)|0\rangle^{\otimes L}$, where $U(\theta)$ is a unitary operation defined by a sequence of quantum gates with tunable parameters θ (e.g., rotation angles).

The optimization process mirrors the training of a classical neural network:

- **Model:** The Quantum Ansatz (the specific arrangement of gates).
- **Weights:** The gate parameters θ .
- **Loss Function:** A physical observable to minimize (e.g., Energy or Infidelity).
- **Optimizer:** A gradient-based algorithm (e.g., Adam) that updates θ to minimize the loss.

The expressibility and trainability of the VQC depend critically on the circuit architecture. We investigate three distinct ansatz designs with varying connectivity and inductive biases. For all experiments, we fix the system size to $L = 4$ (even) to ensure a unique singlet ground state, avoiding Kramer's degeneracy which complicates the spectrum for odd L (this will be discussed later).

Ansatz 1: Hardware Efficient. We take inspiration from Fig. 5.9. in [15], which is designed for the exact same task, finding the ground state of XXZ Hamiltonian. Notice that we have added a first layer of Pauli X gates so our state becomes $|1\rangle$ in all qubits. Afterwards we apply a layers of single-qubit rotations (R_y and R_z) followed by nearest-neighbour CNOT entangling gates, consult Fig. 1.

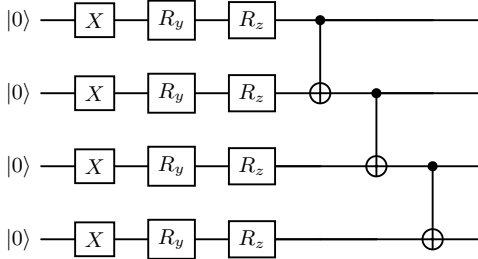


FIG. 1. Schematic of the Ansatz 1 (Hardware Efficient) architecture. The diagram depicts the initial Pauli X gates and subsequently, the repetition block consisting of parameterized single-qubit rotations followed by a ladder of nearest-neighbour CNOT entangling gates.

We expect this circuit to be easy to implement on hardware, but may require greater depth to yield good results.

Ansatz 2: Hamiltonian-Inspired. Ansatz constructed to mimic the time-evolution operator $U(t) = e^{-itH}$. Notice

that the Hamiltonian contains XX, YY, ZZ interactions. So, instead of using CNOTs, we now employ parameterized two-qubit gates to match these terms, inspired by [16]. Our design is shown in Fig. 2.

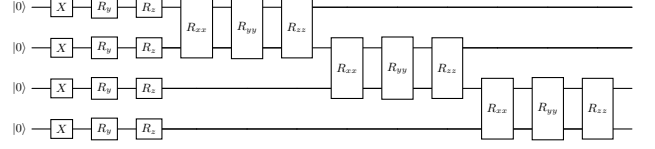


FIG. 2. Schematic of the Ansatz 2 (Hamiltonian-Inspired) architecture. The diagram depicts the initial Pauli X gates and subsequently, the repetition block consisting of parameterized single-qubit rotations followed by a ladder of nearest-neighbour parameterized two-qubit entangling gates.

We expect this ansatz to outperform the others as its structure aligns with the physical symmetries of the target state.

Ansatz 3: All-to-All. This architecture maximizes connectivity, allowing entanglement between any pair of qubits.

A schematic representation of Ansatz 3 is omitted due to the visual complexity of depicting the all-to-all connectivity. Structurally, it follows the same pattern as Fig. 2 but entangling gates are applied between *every* pair of qubits.

This architecture is supposed to be highly expressive, capable of generating volume-law entanglement quickly. However, this massive parameter space leads to barren plateaus, making the circuit exponentially difficult to train as the system size increases.

Contrary to classical intuition, increasing the number of parameters in a VQC does not guarantee superior performance [17]. While highly parameterized circuits like Ansatz 3 possess high expressibility, they often suffer from low trainability. By introducing non-physical, long-range couplings, we create a complex optimization landscape prone to local minima and barren plateaus. Instead of converging these unneeded parameters to zero, the optimizer is likely to get trapped in a state that minimizes energy reasonably well but fails to capture the true state.

We apply these circuits to distinct physical problems.

1. Ground State

To find the ground state $|\psi_0\rangle$, we define the loss function as the expectation value of the Hamiltonian:

$$\mathcal{L}_{\text{VQE}}(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle. \quad (3)$$

By minimizing this quantity via gradient descent, the Variational Principle guarantees convergence towards the true ground state energy E_0 .

2. Time Evolution

Here, the goal is to track the dynamic evolution of the state $|\psi(t)\rangle = e^{-itH} |+\rangle^{\otimes L}$. The target state $|\psi(t)\rangle$ is calculated via ED, we treat this as a supervised learning task. We maximize the quantum fidelity at each time step $t_i \in (0, t_{\max})$, $i = 1, \dots, 10$ and $t_{\max} = 4\pi$:

$$\mathcal{L}_{\text{TE}}(\theta) = 1 - |\langle \psi(\theta) | \psi_{\text{target}}(t_i) \rangle|^2. \quad (4)$$

We compare two initialization strategies:

- **Independent Random Initialization:** Resetting θ randomly at each t_i . Testing the optimizer's ability to find the global minimum from scratch.
- **Sequential “Warm Start” Initialization:** Initializing $\theta(t_i)$ using the optimized parameters from the previous step $\theta(t_{i-1})$. Exploiting the physical continuity of the wavefunction: since $|\psi(t)\rangle$ evolves smoothly, the optimal parameters should also traverse a smooth trajectory in parameter space. This strategy is expected to yield higher stability [18].

3. Excited States

To find the first excited state $|\psi_1\rangle$, we employ Variational Quantum Deflation (VQD). We modify the cost function to minimize energy while enforcing orthogonality to the previously found ground state $|\psi_0\rangle$:

$$\mathcal{L}_{\text{VQD}}(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle + \beta |\langle \psi(\theta) | \psi_0 \rangle|^2. \quad (5)$$

The penalty term ($\beta > 0$) effectively “masks” the ground state, creating a new global minimum at the first excited energy level.

4. Kramers’ Degeneracy

At zero magnetic field, the XXZ Hamiltonian preserves Time-Reversal Symmetry. For systems with an odd number of half-integer spins, Kramers’ Theorem dictates that the total spin is half-integer, rendering every eigenstate, including the ground state, doubly degenerate. This creates a critical distinction for VQD:

- Even L (e.g., $L = 4$): The ground state is a unique singlet ($S_{\text{tot}} = 0$), so VQD functions normally.
- Odd L (e.g., $L = 5$): The ground state is a degenerate doublet. Consequently, standard VQD fails to find the true first excited state; it instead identifies the orthogonal vector within the ground subspace, effectively measuring a zero gap ($E_1 - E_0 = 0$).

This poses a fundamental problem for the standard VQD algorithm. If we only penalize the overlap with a single ground state vector $|\psi_0\rangle$, the optimizer will simply converge to its degenerate orthogonal partner $|\psi_1\rangle$ (where $E_1 = E_0$), rather than the first physical excitation (E_2).

We implement a Generalized Subspace VQD: instead of orthogonalizing against a single state, we first identify the entire ground-state subspace $S_{GS} = \{|\psi_k\rangle\}$ via exact diagonalization. The loss function is modified to penalize overlaps with all basis vectors of this subspace:

$$\mathcal{L}_{\text{Gen-VQD}}(\theta) = \langle \hat{H} \rangle + \beta \sum_{k \in S_{GS}} |\langle \psi(\theta) | \phi_k \rangle|^2. \quad (6)$$

This forces the variational ansatz to “jump” over the entire ground-state manifold and converge to the true first excited energy level (E_2), effectively recovering the physical spectral gap.

IV. RESULTS AND DISCUSSION

This section presents the numerical results obtained from our simulations, our code can be found in [19]. We begin by establishing the ED benchmarks, which serve as the ground truth for evaluating the variational methods.

Subsequently, we analyze the performance of the proposed VQC across distinct tasks: ground state preparation, real-time evolution and excited state search.

A. Exact Diagonalization

We begin by establishing the classical benchmark for our study. In this section, we analyze the ground state properties of the XXZ chain as a function of the anisotropy parameter $\delta \in [-2, 2]$ and system size $L = 2, 3, \dots, 10$ (“small” enough so that we can brute force diagonalise). These results serve as the “ground truth” for evaluating the variational methods in subsequent sections.

a. Ground State Energy.

Fig. 3 displays the ground state energy E_0 for different values of δ and L .

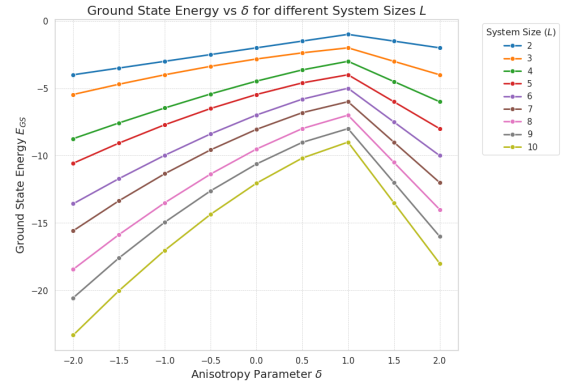


FIG. 3. Ground state energy for different values of δ and L . Results obtained via exact diagonalization.

Two distinct physical behaviours are immediately apparent, separated by the critical point at $\delta = 1$:

- $\delta < 1$ (XY / Antiferromagnetic Phase): The energy varies smoothly with δ , characteristic of the gapless critical phase (or the antiferromagnetic phase for $\delta < -1$).
- $\delta > 1$ (Ferromagnetic Ising Phase): The system undergoes a first-order phase transition. The energy becomes linear in δ , minimizing the interaction term $-\delta \sum \hat{Z}_i \hat{Z}_{i+1}$ by aligning all spins.

Additionally, we observe the extensivity of the energy, where E_0 scales linearly with system size L (lower total energy for larger systems).

b. Entanglement Entropy.

The bipartite entanglement entropy serves as a proxy for the complexity of the quantum state. Fig. 4 illustrates the von Neumann entropy for a half-chain partition.

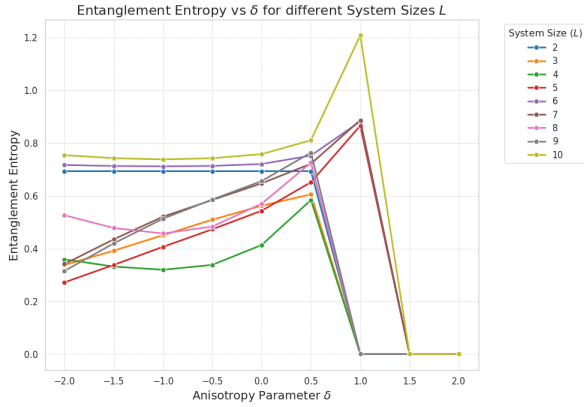


FIG. 4. Entanglement entropy for different values of δ and L . Results obtained via exact diagonalization.

Since the chosen partition was half of the spin chain, i.e. $\min(d_A, d_B) = 2^{L/2}$ we verified that all results satisfy the theoretical bound given by Eq. (2). The behaviour strongly correlates with the phase transition:

- Critical Region ($\delta < 1$): The system exhibits significant entanglement, indicating long-range quantum correlations. This confirms that the ground state in the XY phase is highly non-trivial and cannot be represented by a simple product state.
- Ferromagnetic Region ($\delta > 1$): The entropy collapses rapidly to zero. The ground state approaches the product state $|\downarrow\downarrow\ldots\downarrow\rangle$ (or $|\uparrow\uparrow\ldots\uparrow\rangle$), which is separable and thus has zero entanglement. However, due to this degeneracy, a superposition of these two product states $|\psi\rangle = \alpha|\uparrow\uparrow\ldots\uparrow\rangle + \beta|\downarrow\downarrow\ldots\downarrow\rangle$, also yields the minimal ground state energy. In this case, the entanglement entropy would not be zero.

We also observe distinct parity effects, where odd and even L systems display slightly different entropy profiles, a precursor to the degeneracy issues discussed later.

c. Magnetization and Variances.

Finally, we analyze the order parameters via the total Pauli operators $\hat{\tau}^\alpha = \sum_i \hat{\sigma}_i^\alpha$. The expectation values and variances are presented in Fig. 5.

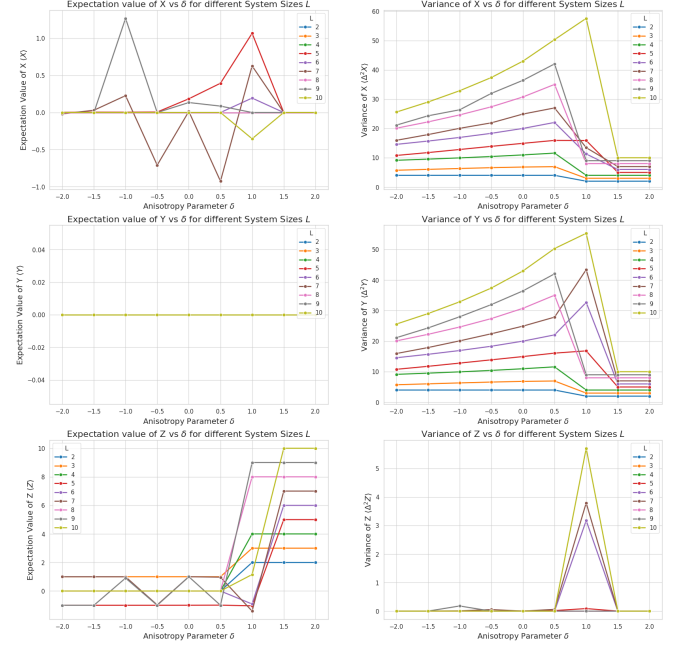


FIG. 5. Expectation values (left) and variances (right) of the total Pauli operators for different values of δ and L . Results obtained via exact diagonalization.

Regarding the expectation values, we can safely say that our results are consistent with the $U(1)$ symmetry of the Hamiltonian: the transverse magnetizations vanish ($\langle \hat{X} \rangle = \langle \hat{Y} \rangle \approx 0$) for all δ . We attribute the non-zero fluctuations to numerical precision.

Moreover, the longitudinal magnetization $\langle \hat{Z} \rangle$ serves as the order parameter: for $\delta < 1$, the system resides in the $\langle \hat{Z} \rangle = 0$ sector (antiferromagnetic/XY phase) whereas at $\delta > 1$ we observe a sharp transition to saturation $\langle \hat{Z} \rangle = L$, confirming the onset of the Ferromagnetic phase where all spins align.

Furthermore, the variances $\langle \Delta^2 \hat{\tau} \rangle$ provide insight into quantum fluctuations. For $\langle \Delta^2 X \rangle$ and $\langle \Delta^2 Y \rangle$, the variance grows as $\delta \rightarrow 1$, peaking at the critical point. This reflects the intense quantum fluctuations associated with the phase transition. Notice that for $\delta > 1$, the variances drop to zero, consistent with a product state with fixed spin orientation.

Our results: $\langle \Delta^2 Z \rangle = 0$, match the conservation of total magnetization previously mentioned. Some discrepancies are found at $\delta = 1$, due to the fact that at the critical point the spectral gap closes and the ground state degeneracy increases. The non-zero variance reflects numerical mixing within this degenerate subspace, highlighting the sensitivity of the eigensolver at the critical point.

In summary, the Exact Diagonalization results faithfully reproduce the known physics of the XXZ model, providing a robust baseline for validating our variational quantum circuits.

B. Variational Quantum Circuits

We now present the VQC simulation results. To ensure a focused analysis, we *fixe* the system size to $L = 4$. In the following figures, the colour coding is consistent: blue represents Ansatz 1 (Hardware Efficient), orange corresponds to Ansatz 2 (Hamiltonian-Inspired), and green denotes Ansatz 3 (All-to-All).

1. Ground State

Our primary objective is to find the ground state of the Hamiltonian by minimising Eq. (3). Fig. 6 displays the fidelity $F = |\langle \psi_{VQC} | \psi_{ED} \rangle|^2$ relative to the exact benchmark.

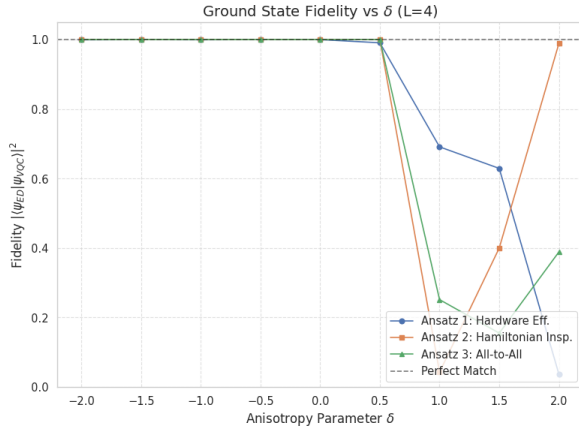


FIG. 6. Fidelity between VQC and ED ground states for $L = 4$ versus anisotropy δ .

For the XY phase ($\delta < 1$), all three models achieve near-perfect fidelity ($F \approx 1.0$), indicating successful convergence to the unique ground state. However, performance degrades noticeably upon entering the ferromagnetic phase ($\delta > 1$). This discrepancy arises from the ground state degeneracy of the Hamiltonian in this region: the VQC finds a minimum energy state which does not correspond to the one the ED returns.

Ansatz 2 (Hamiltonian-Inspired) demonstrates superior performance, recovering a fidelity $F \approx 1$ when $\delta = 2$. This validates our hypothesis that embedding the physical symmetries of the Hamiltonian into the circuit significantly enhances trainability.

To further investigate, we now compare the bipartite entanglement entropy, shown in Fig. 7.

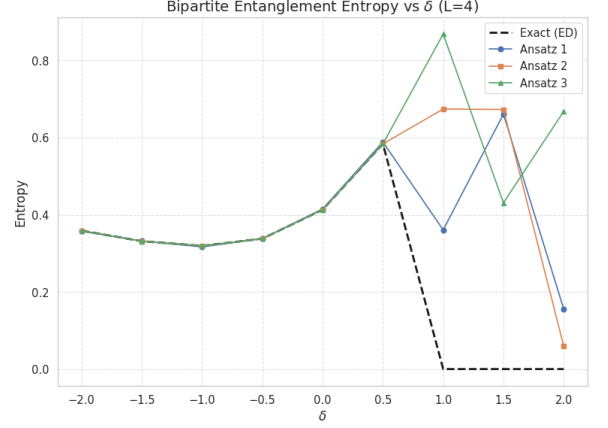


FIG. 7. Bipartite entanglement entropy of VQC and ED ground states for $L = 4$ versus anisotropy δ .

The entropy results mirror the fidelity trends. In the gapless phase ($\delta < 1$), all circuits reproduce perfectly the system's entanglement. However, a significant deviation occurs for $\delta \geq 1$. The ground state should correspond to a separable product state, but the VQC converges to a superposition of these product states, yielding a non-zero entropy value. Ansatz 2 again provides the closest approximation to the physical reality away from the transition point. On the other hand, Ansatz 3 fails drastically; the optimizer has found a state that minimizes the energy but is highly entangled, a result of over-parameterization.

Finally, we examine the order parameters in Fig. 8.

The results are surprisingly accurate; for the x and y directions the magnetization remains zero for all values of δ apart from $\delta = 1$. For the z direction all circuits adapt perfectly to the ED line until reaching the transition point. Once again, the Hamiltonian Inspired ansatz yields the best approximation.

Our whole study highlights the critical role of circuit architecture. Ansatz 2 (Hamiltonian-Inspired) consistently outperforms. Ansatz 3's (All-to-All) poor performance suggests that for local Hamiltonians, more connectivity is not necessarily better; rather, the circuit structure should reflect the locality of the physical interactions. Ansatz 1 (Hardware Efficient) performs adequately despite using non-parameterized entanglers (CNOTs), likely due to the flexibility provided by its single-qubit rotation layers.

A consistent trend emerges across all properties: VQCs struggle to capture the sharp physics of the quantum phase transition. While the ansatzes perform excellently in the smooth Antiferromagnetic/XY phase, we observe a systematic degradation in fidelity and accuracy near the critical point and into the Ferromagnetic phase. The poor performance in this region is an artifact of the ground state degeneracy.

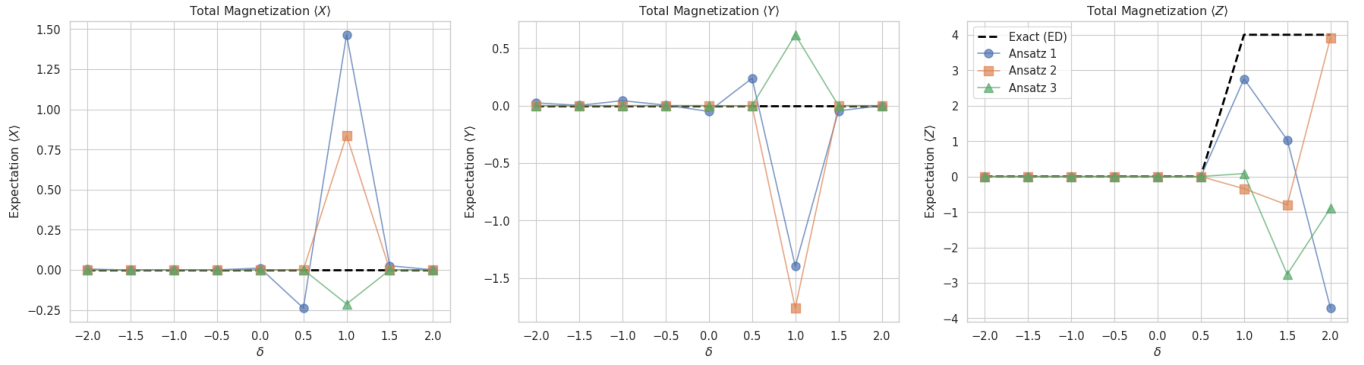


FIG. 8. Magnetization order parameters for VQC and ED ground states for $L = 4$ versus δ .

2. Time-Evolution

We extend our analysis to the dynamic simulation of the system. We will compare the two different initialization strategies when minimizing Eq. (4). It is worth noting that in this section, our loss function compares the *fidelity* between two states.

The results obtained with independent random initialization are displayed in Fig. 9.

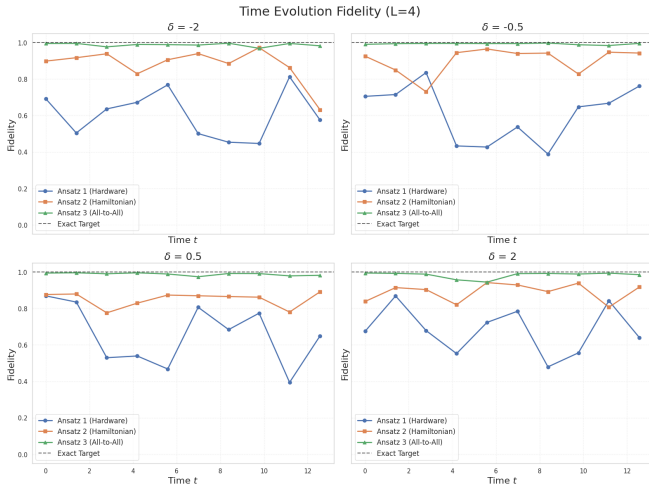


FIG. 9. Time evolution of fidelity using independent random initialization for varying δ .

The ansatzes maintain a fidelity $F > 0.4$ throughout all times and δ values, however the performance is characterized by some fluctuations. The optimizer struggles to consistently locate the global minimum when starting from scratch.

Alternatively, Fig. 10 show the results obtained with the sequential “warm-start” initialization.

This strategy yields a marked improvement. All ansatzes maintain a higher fidelity with high stability across the entire simulation window.

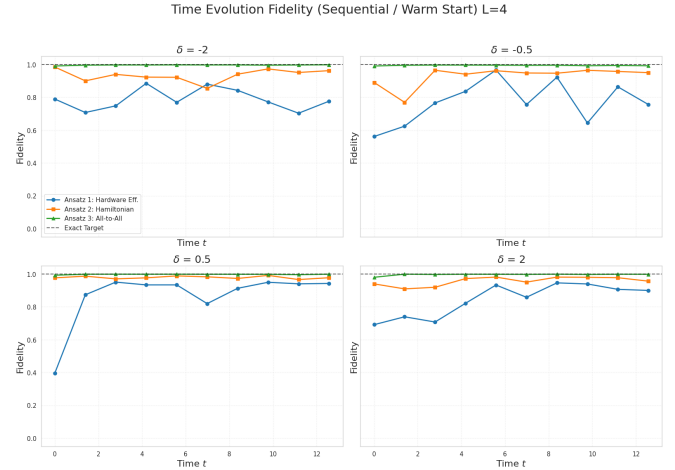


FIG. 10. Time evolution of fidelity using sequential “warm-start” for varying δ .

Something worth noting are the extraordinary results from Ansatz 3 in both initialization strategies. While the high parameter count creates optimization challenges for finding the ground state when minimising the energy, it simultaneously grants the circuit the high expressibility required to represent the target quantum state perfectly. This high fidelity is a direct result of over-parameterization, which allows the ansatz to effectively cover the entire Hilbert space for small system sizes, ensuring that an exact solution exists within the reachable parameter space.

Furthermore, this result highlights a critical distinction in optimization objectives: when the loss function is set to maximize fidelity, the circuit’s massive capacity becomes an asset. This capacity to overfit renders any comparison of initialization strategies inconclusive.

Therefore, focusing on Ansatz 1 and 2, the results demonstrate that parameter initialization is a decisive factor. The sequential approach acts as a form of transfer learning, drastically reducing the optimization complexity.

3. First Excited State

We now determine the first excited state $|\psi_1\rangle$ by minimizing Eq. (5), which penalizes overlap with the previously determined ground state. We benchmark these results against the exact spectrum.

a. Energy Spectrum.

Fig. 11 compares the VQC optimized energy with the exact first excited energy E_1 .

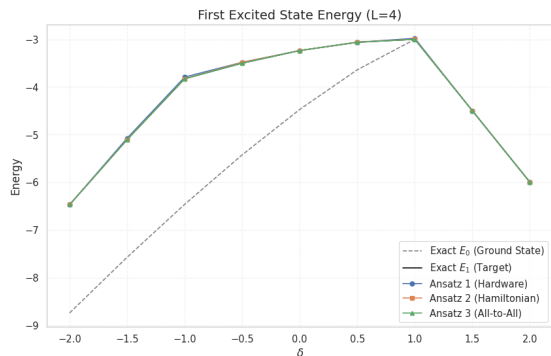


FIG. 11. First excited state energy E_1 of VQC and ED for $L = 4$ versus anisotropy δ .

There's an almost perfect agreement with the exact solution across the entire domain. Note that in the ferromagnetic phase the energy gap closes and E_1 converges to E_0 , proving the degeneracy of the ground state when $\delta > 1$, we have been mentioning. The VQD algorithm successfully identifies this degeneracy, finding an orthogonal state with an energy nearly identical to the ground state.

b. Fidelity Analysis.

The fidelity of the excited state is presented in Fig. 12.

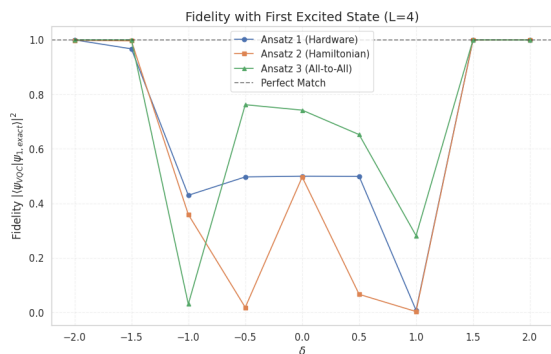


FIG. 12. Fidelity between VQC and ED first excited states for $L = 4$ versus anisotropy δ .

The accuracy degrades significantly near the critical points, where the closing of the spectral gap and the divergence of the correlation length make the optimization landscape particularly challenging.

c. Magnetization and Variances.

Finally, we verify the physical properties of the excited state by inspecting the magnetization in Fig. 13.

The results are fairly satisfactory. The transverse magnetizations ($\langle X \rangle, \langle Y \rangle$) vanish, consistent with symmetry requirements. The longitudinal magnetization $\langle Z \rangle$ is not captured accurately.

4. Kramers' Degeneracy

A critical physical anomaly emerged when extending our analysis to odd system sizes ($L = 5$). Initial simulations using the standard VQD algorithm failed to identify a finite energy gap, instead returning an excited energy $E_1 = E_0$ a clear manifestation of Kramers' Degeneracy.

Fig. 14 shows the failure of the standard approach, where the VQC results collapse onto the ground state line (not visible due to overlap). To resolve this we changed our loss function to Eq. (6), penalizing overlaps with the entire ground-state subspace. Fig. 15 demonstrates the success of our Generalized VQD.

It is worth noting that we employed Ansatz 3 which proved sufficiently to capture the excitation once the loss landscape was corrected. This generalizes our solver to handle any system size L , automatically detecting and avoiding symmetry-protected subspaces.

V. OUTLOOK

The main conclusion drawn from this study is the outperformance of Ansatz 2, which highlights the importance of designing the circuit to match the Hamiltonian.

The All-to-All connectivity yields worse results than expected due to over-parameterization; however, it achieves perfect results when comparing state fidelities.

Future work could investigate other architectures. Variations of Ansatz 2 might yield faster and superior results, particularly for approximating behaviour near the phase transition, which should be a priority.

Additionally, parameters such as learning rate and circuit depth were kept constant, varying these in future studies would provide better insight into VQC performance.

Finally, a robust analysis of the physical system is crucial. Blindly applying VQC algorithms to systems with degeneracies (like odd-sized chains or the ground state of the ferromagnetic phase) fails, requiring physical insight to correct the algorithm. This study could be extended to higher excited states, as well as addressing the instability in the ferromagnetic phase, a possible solution would be to add a small field to the Hamiltonian to break the degeneracy.

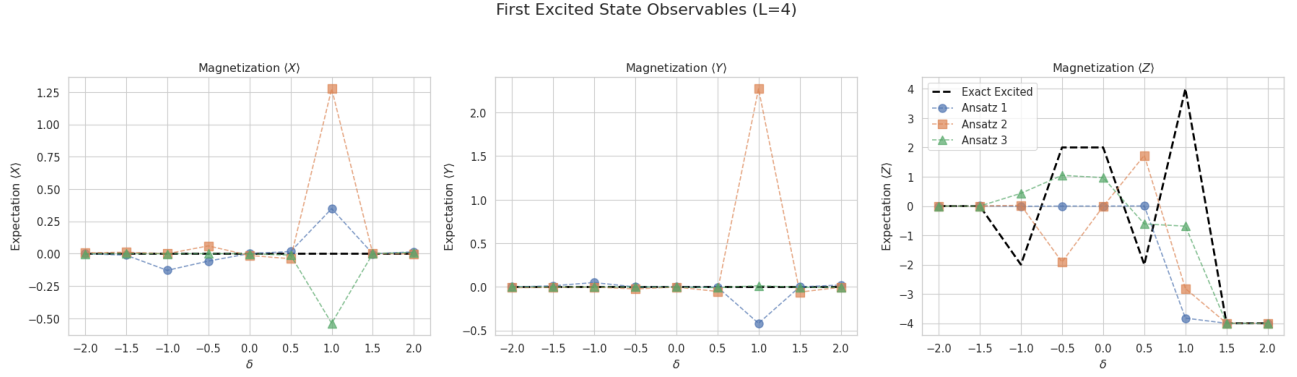


FIG. 13. Magnetization of VQC and ED first excited states for $L = 4$ versus anisotropy δ .

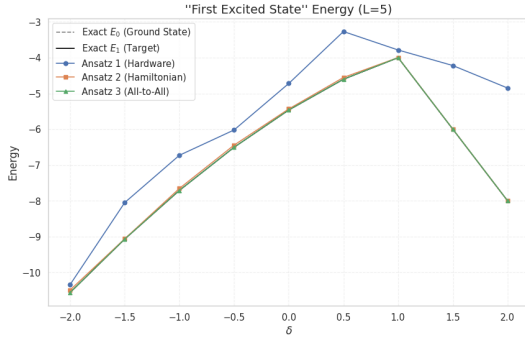


FIG. 14. Failure of the standard VQD algorithm for $L = 5$ (odd). The VQC optimization converges to the degenerate ground state partner E_0 , due to the lack of orthogonality constraints against the full ground-state subspace.

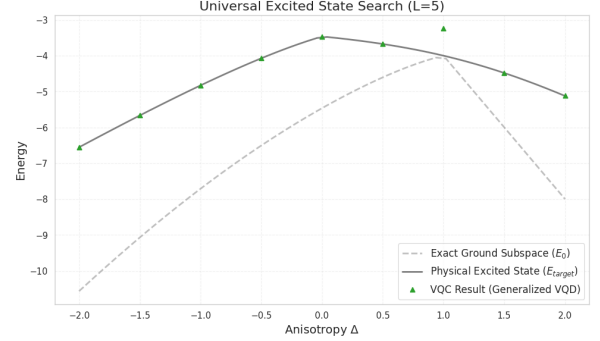


FIG. 15. Excited state search for $L = 5$ (odd). The VQC (points) successfully skips the degeneracy of the ground subspace and converges to the first physical excitation. Note that Ansatz 3 (All-to-All) was used for this demonstration.

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